

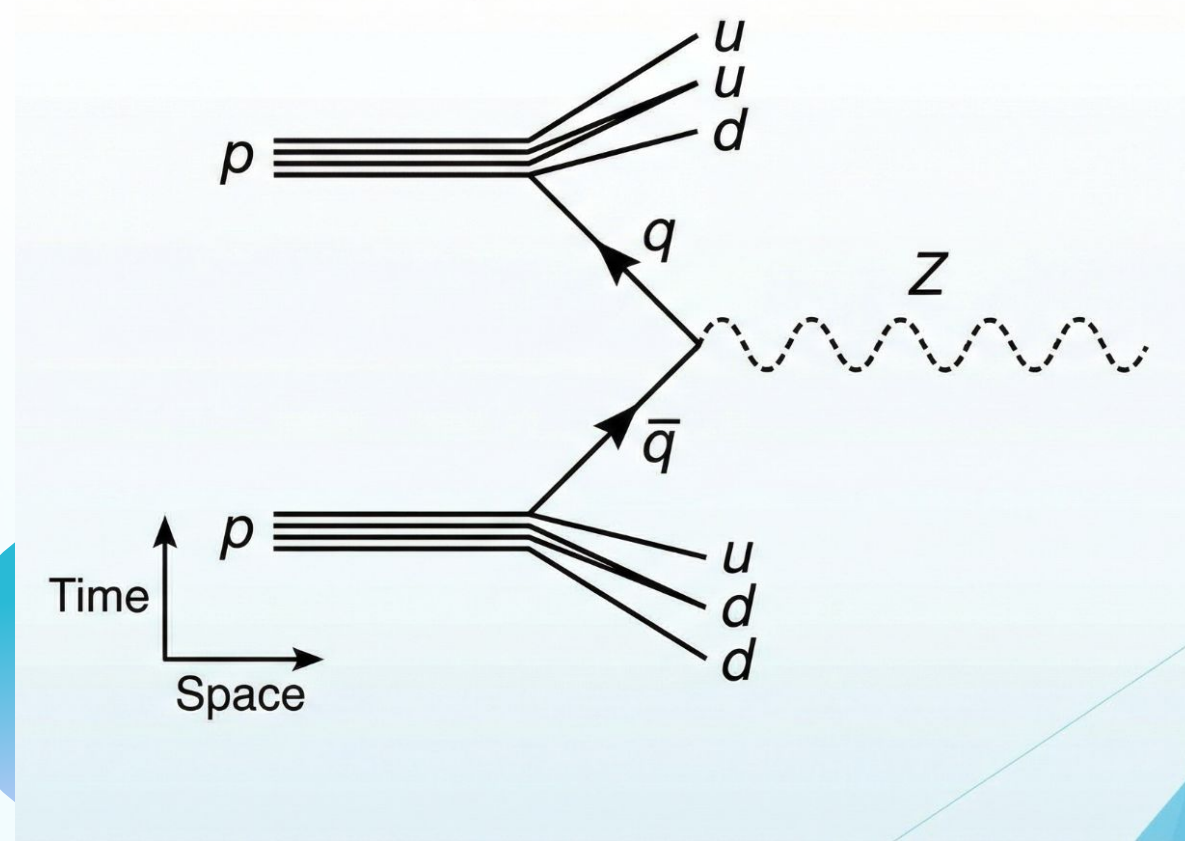
A Parton-Level SMEFT Analysis of Diboson Production at the LHC

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The Problem

Why Study Diboson Production with SMEFT?

The Challenge:

- The Standard Model is incomplete which means that new physics exists at higher energy scales.
- Direct searches have found nothing yet which raises the need for model-independent approaches.

The Approach:

SMEFT (Standard Model Effective Field Theory) parameterizes unknown new physics through higher-dimensional operators.

Diboson production (WW, WZ, ZZ) probes electroweak interactions sensitive to these operators

The Question here is:

Can machine learning detect subtle EFT-induced shape changes in kinematic distributions?



Computational Approach

Event Generation & EFT Framework

- Process: $pp \rightarrow ZZ \rightarrow \ell^+ \ell^- \nu \bar{\nu}$ at LO
Generator: MadGraph5_aMC@NLO v2.9.26
EFT Model: SMEFTsim (U35_MwScheme)
Analysis: Python + uproot + scikit-learn

Method

- Generated SM events with embedded reweighting for 9 Wilson coefficients (WC's)
- WC's: $c_W, c_{HW}, c_{H\Box}, c_{Hq3}, c_{qq1}, c_{qq11}, c_{qq3}, c_{qq31}, c_{ll1}$
- Each event carries 19 weights: SM + (± 1 for each operator)
- The simulation enables efficient parameter space exploration without regeneration

LO: Leading Order

Analysis:

From Events to Machine Learning

- MadGraph $\rightarrow$ LHE $\rightarrow$ ROOT $\rightarrow$ Python Analysis $\rightarrow$ ML Training

Kinematic Features Constructed:

- Leading & trailing lepton p_T
- Dilepton system p_T (Z proxy)
- Missing transverse energy (MET)
- Lepton pseudorapidities (η)

While establishing this, parton showers, detector simulation, and higher-order corrections were excluded so that only pure matrix-element effects were considered.

Goal: Test intrinsic EFT sensitivity and maintain clean LO interpretation

Results

Operator Survey

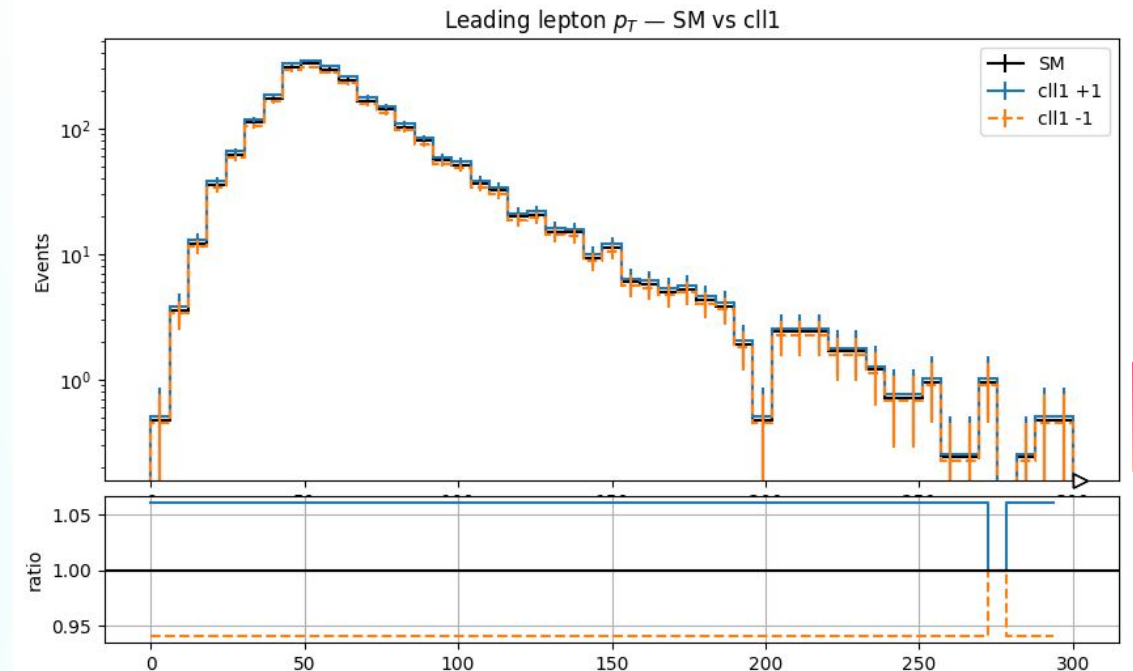
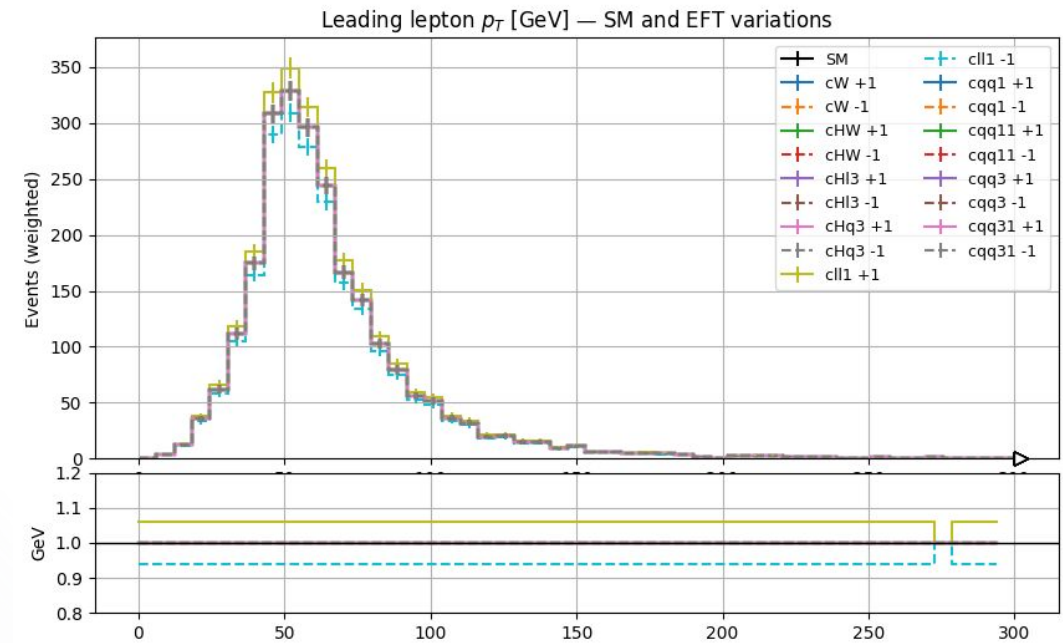
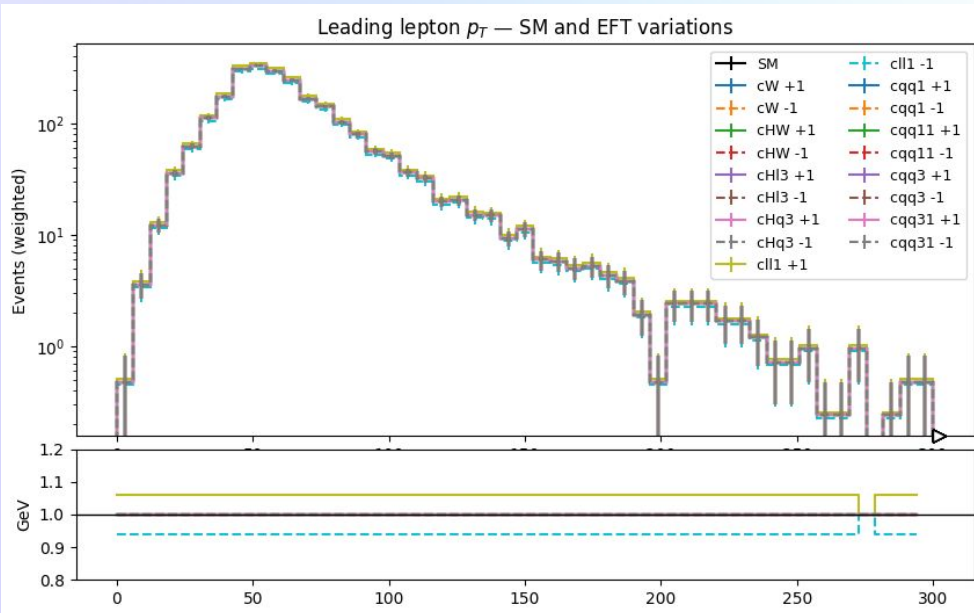
Which Operators Affect This Process?

- Out of 9 operators tested, only $cll1$ produces non-trivial reweighting effect on ZZ .
- cW , cHW , $cHl3$, $cHq3$ results in no effect (weights = SM)
- $cqq1$, $cqq11$, $cqq3$, $cqq31$ also showed no effect
- Only $cll1$ showed $\sim 6\%$ weight modification

All operators except $cll1$: flat ratio at 1.0

$c\Box\Box_1$: flat ratio at 1.06 ($C=+1$) or 0.94 ($C=-1$)

Observation: The ratio is constant across all kinematic variables.



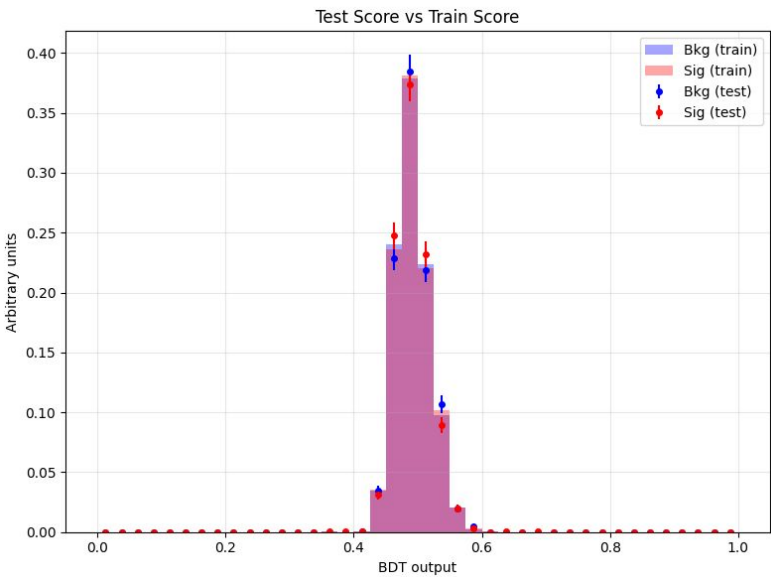
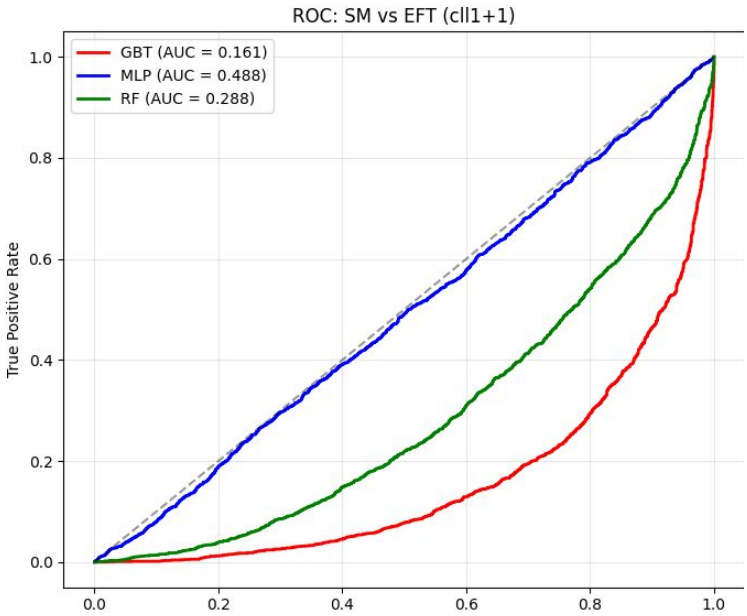
Results

Machine Learning

Can ML Distinguish SM from EFT?

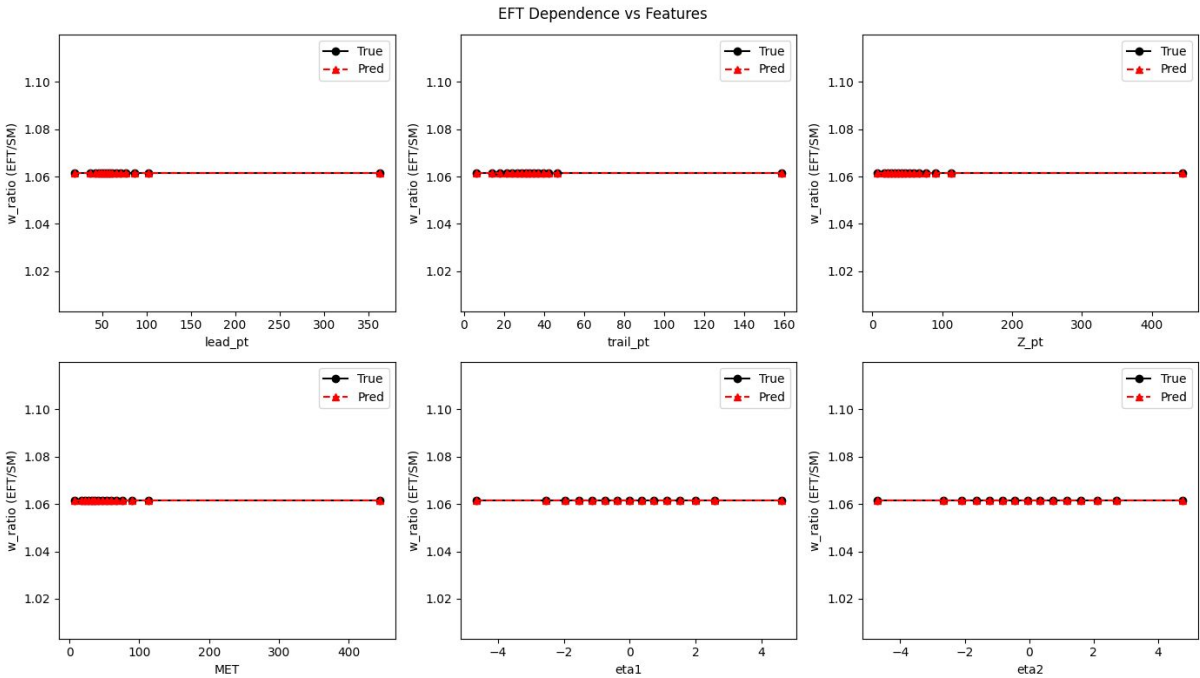
Classification Task: SM (label=0) vs EFT cll1 (label=1)

Model	Area Under Curve	Interpretation
Gradient Boosted Trees	0.16	No signal to learn
MLP (2-layer)	0.49	Random guessing
Random Forest	0.29	No signal to learn



Regression Task: Predict weight ratio from kinematics

Model	R ²	Interpretation
Gradient Boosted Trees	1.00	Learned the constant (1.06)
MLP/RF	0.00	No variance to explain



Conclusion: ML correctly identifies that no kinematic discrimination exists

Interpretation of Results

From this analysis, I was able to understand that the cll operator just scales all the vertices with respect to the SM in the tree level diagrams. This verified the physics of the operator.

The Operator

$$\mathcal{O}_{ll}(1) = (\bar{\ell} \gamma_{\mu} \ell)(\bar{\ell} \gamma_{\mu} \ell)$$

Insight obtained

ClI produces normalization-only effects in $pp \rightarrow ZZ$

Shape-based ML discrimination is fundamentally impossible for this operator

Sensitivity requires rate measurements, not differential distributions

Contrast with other operators:

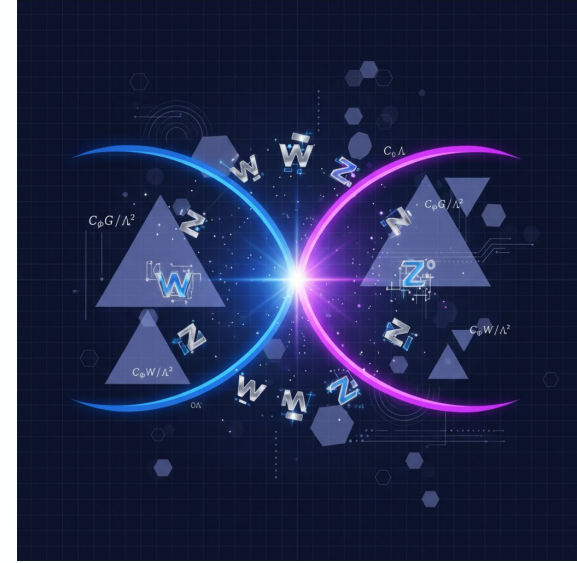
- Operators like cW modify triple gauge couplings
- They introduce p_T -dependent effects visible in high-energy tails
- But cW doesn't contribute to ZZ at tree level

What This Analysis Achieved

- Event generation with EFT reweighting
- Kinematic reconstruction
- ML classification and regression
- SHAP interpretability analysis

Physics

- This is a 4-lepton contact interaction
- It modifies the total cross-section only and does not introduce energy-enhanced terms (no $\hat{s}/\Lambda^2$ growth)
- Kinematic distributions remain SM-shaped



THANK YOU

